

Netflix Dataset Analysis Report

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This project focused on the cleaning and analysis of a Netflix content dataset using Python and Power BI to identify data quality issues and generate insights into content distribution, ratings, and trends. The objective was to identify and resolve data quality issues, structure the dataset for reliable analysis, and generate meaningful insights into Netflix's content distribution, ratings, release trends, and viewing patterns through exploratory data analysis.

The dataset contained 8,808 records and included key attributes such as type, title, director, country, release year, rating, and duration. During familiarization, dataset structure and data types were reviewed to identify numerical, categorical, and date-based variables. Release year and duration were treated as numerical features, while type, country, rating, and genre were classified as categorical variables. The dateadded column was converted to datetime format for time-series analysis.

Several data cleaning procedures were performed to improve data quality. Missing values in director, cast, country, and rating were handled using appropriate placeholders. The duration column was split into minutes and seasons, and text fields and date formats were standardized for consistency. Rating values such as "NR" and "Not Rated" were unified for accuracy. No duplicate records were found in the dataset.

Exploratory Data Analysis was performed using summary statistics and visualizations to identify patterns in ratings, release years, durations, content types, countries, and genres. Additional analysis included content type distribution, yearly content additions, top content-producing countries, and genre concentration.

Visualizations such as bar charts, histograms, and time-series plots were used to explore distribution, trends, and relationships in the dataset. A content distribution chart compared Movies and TV Shows, while a yearly additions chart examined Netflix's growth over time. Country-level analysis highlighted leading content producers, while ratings and release year visualizations revealed audience targeting and content concentration patterns.

Key business metrics analyzed included Total Titles, Total Movies, Total TV Shows, and Average Movie Duration. The analysis showed that movies dominate the catalog, TV-MA and TV-14 are the most common ratings, and content additions increased significantly after 2015, reflecting rapid platform expansion.

The project was completed using Python (Pandas, Matplotlib, Seaborn), Jupyter Notebook, and Power BI for visualization and dashboard development.